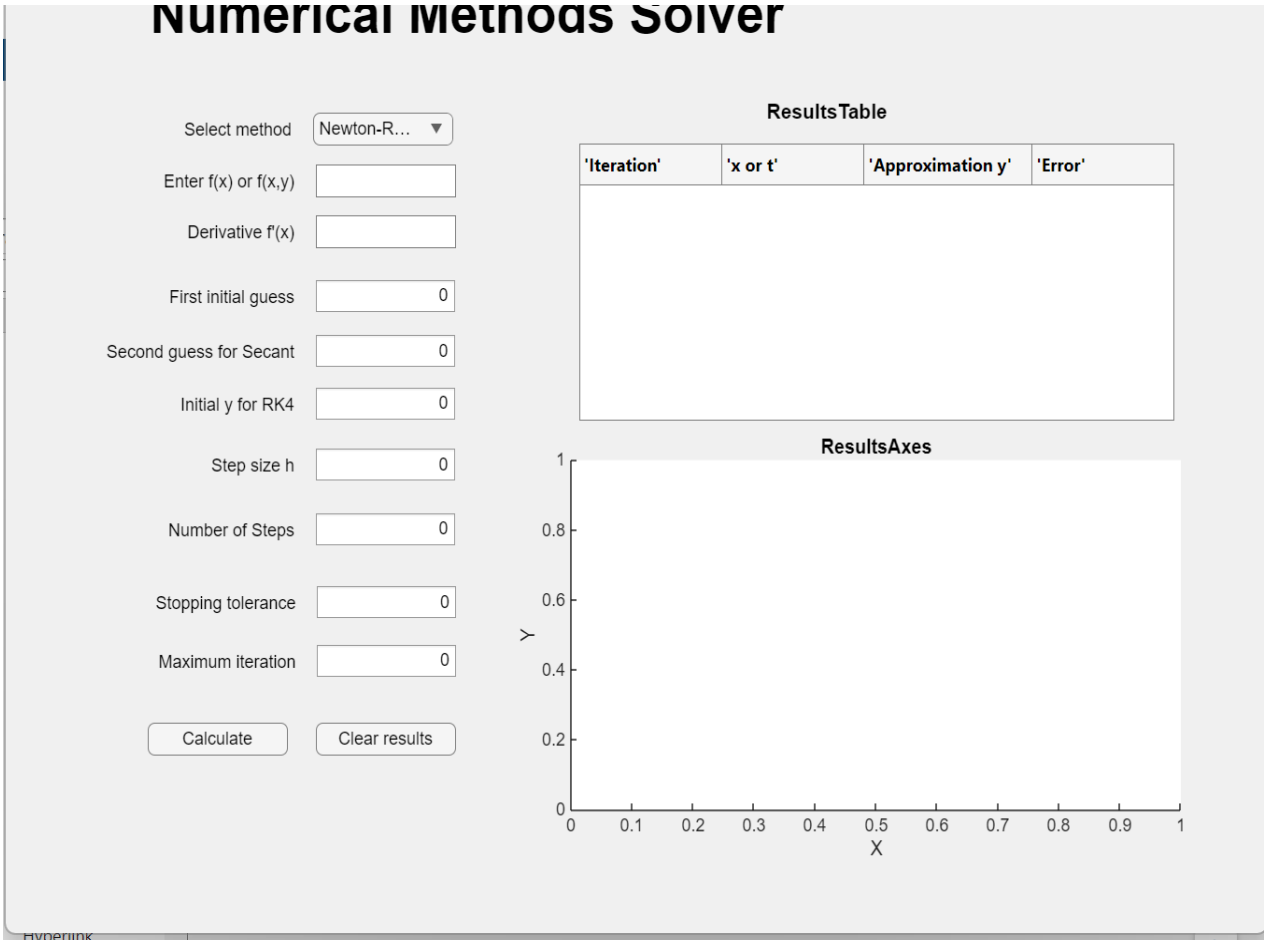
# APP BUILDING

## Questions

**Design an app that can solve numerical methods(secant,rahpson and runge kutta).**

**DESIGN VIEW:**



**CODE VIEW:**

```
classdef NumericalMethodsSolver_Backup < matlab.apps.AppBase


    % Properties that correspond to app components
    properties (Access = public)
        UIFigure                    matlab.ui.Figure
        NumberofStepsEditField      matlab.ui.control.NumericEditField
        NumberofStepsEditFieldLabel matlab.ui.control.Label
        ResultsTableLabel           matlab.ui.control.Label
        DerivativefxEditField_2     matlab.ui.control.EditField
        DerivativefxEditField_2Label matlab.ui.control.Label
        DerivativefxEditField       matlab.ui.control.EditField
        DerivativefxEditFieldLabel  matlab.ui.control.Label
        EnterfxorfxyEditField       matlab.ui.control.EditField
        EnterfxorfxyEditFieldLabel  matlab.ui.control.Label
        UITable                     matlab.ui.control.Table
        ClearresultsButton          matlab.ui.control.Button
        CalculateButton             matlab.ui.control.Button
        MaximumiterationEditField   matlab.ui.control.NumericEditField
        MaximumiterationEditFieldLabel  matlab.ui.control.Label
        StoppingtoleranceEditField  matlab.ui.control.NumericEditField
        StoppingtoleranceEditFieldLabel  matlab.ui.control.Label
        StepsizehEditField          matlab.ui.control.NumericEditField
        StepsizehEditFieldLabel     matlab.ui.control.Label
        InitialyforRK4EditField     matlab.ui.control.NumericEditField
        InitialyforRK4EditFieldLabel  matlab.ui.control.Label
```

```
    SecondguessforSecantEditField   matlab.ui.control.NumericEditField
    SecondguessforSecantEditFieldLabel  matlab.ui.control.Label
    FirstinitialguessEditField     matlab.ui.control.NumericEditField
    FirstinitialguessEditFieldLabel  matlab.ui.control.Label
    SelectmethodDropDown          matlab.ui.control.DropDown
    SelectmethodDropDownLabel      matlab.ui.control.Label
    NumericalMethodsSolverLabel    matlab.ui.control.Label
    UIAxes                  matlab.ui.control.UIAxes
end

% Callbacks that handle component events
methods (Access = private)

    % Button pushed function: CalculateButton
    function CalculateButtonPushed(app, event)

        method = strtrim(char(app.SelectmethodDropDown.Value));
        equation = strtrim(char(app.EnterfxorfxyEditField.Value));
        isNewton = strcmpi(method,"Newton-Raphson");
        isSecant = strcmpi(method,"Secant");
        isRK4 = strcmpi(method,"Runge-Kutta") || contains(method,"rk4");
        if ~isNewton && ~isSecant && ~isRK4
            uialert(app.UIFigure,'Please select a valid numerical method.',"Input Error");
            return;
        end
        if isempty(equation)
```

```
        uialert(app.UIFigure,"Please enter the equation","Input Error");
        return;
    end
    try
        if isNewton
    x0 = app.FirstinitialguessEditField.Value;
    tol = app.StoppingtoleranceEditField.Value;
    maxIter = floor(app.MaximumiterationEditField.Value);
    derivative = strtrim(char(app.DerivativefxEditField_2.Value));
    if isNewton && (~isscalar(x0) || ~isfinite(x0) || ~isscalar(tol)|| tol<=0 ||
~isscalar(maxIter) || ~isfinite(maxIter) || maxIter<1 || maxIter ~=floor(maxIter) ||
isempty(derivative))
        uialert(app.UIFigure,"Check the initial guess, derivative, tolerance and
maximum iterations.","Input Error");
        return;
    end


        f = str2func(['@(x)' equation]);
        df = str2func(['@(x)' derivative]);
       data = zeros(maxIter,4);
        for k = 1:maxIter
           fx = f(x0);
           dfx = df(x0);
           if ~isfinite(fx) || ~isfinite(dfx) || dfx ==0
              error("Invalid function or zero derivative encountered.");
           end
           x1 = x0- fx/dfx;
```

```
            err = abs(x1 - x0);
            data(k,:) = [k,x1,f(x1),err];
            if err<=tol
              break;
            end
            x0 = x1;
          end
            graphX = data(:,1);
            graphY = data(:,2);
            xLabelText = "Iteration";
            yLabelText = "Root Estimate";
            graphTitle = "Newton-Raphson Convergence";
            finalMessage = sprintf("Newton-Raphson converged successfully.\nFinal
root = %.8f\nIterations = %d",x1,k);
          elseif isSecant
            x0 = app.FirstinitialguessEditField.Value;
            x1 = app.SecondguessforSecantEditField.Value;
            tol = app.StoppingtoleranceEditField.Value;
            maxIter = app.MaximumiterationEditField.Value;
            if  ~isscalar(x0) || ~isfinite(x0) || ~isscalar(x1) || ~isfinite(x1) || x0 ==x1 ||
~isscalar(tol) || tol<=0 || ~isscalar(maxIter) || ~isfinite(maxIter) || maxIter<1 || maxIter
~=floor(maxIter)
              uialert(app.UIFigure,"Check both initial guesses, tolerance and maximum
iterations.","Input Error");
              return;
            end
            f = str2func(['@(x)' equation]);
```

```
            data = zeros(maxIter,4);
            for k = 1:maxIter
              f0 = f(x0);
              f1 = f(x1);
denominator = f1 - f0;
if ~isfinite(f0) || ~isfinite(f1)||denominator == 0
error("Secant method encountered an invalid denominator.");
             end
              x2 = x1 - f1*(x1 - x0)/denominator;
              err = abs(x2 - x1);
              data(k,:) = [k,x2,f(x2),err];
              if err <= tol
                break;
              end
              x0 = x1;
              x1 = x2;
            end
                data = data(1:k,:);
                graphX = data(:,1);
                graphY = data(:,2);
                graphTitle = "Secant Method Convergence";
                xLabelText = "Iteration";
                yLabelText = "Root estimate";
                finalMessage = sprintf("Secant method completed successfully.\nFinal
root = %.8f\nInterations = %d",x2,k);
          elseif isRK4
```

```
            equation = strtrim(char(app.EnterfxorfxyEditField.Value));
            y0 = app.InitialyforRK4EditField.Value;
            h = app.StepsizehEditField.Value;
            n = app.NumberofStepsEditField.Value;
            disp(equation);
            disp(y0);
            disp(h);
            disp(n);
            disp(['equation empty: ',num2str(isempty(equation))]);
            disp(['y0 scalar: ',num2str(isscalar(y0))]);
            disp(['y0 finite: ',num2str(isfinite(y0))]);
            disp(['h scalar: ',num2str(isscalar(h))]);
            disp(['hfinite: ',num2str(isfinite(h))]);
            disp(['h<=0: ', num2str(h<=0)]);
            disp(['n scalar: ',num2str(isscalar(n))]);
            disp(['n finite: ',num2str(isfinite(n))]);
            disp(['n<1: ',num2str(n<1)]);
            disp(['n not integer: ',num2str(n~= floor(n))]);
            disp('REACHED RK4 VALIDATION');
            disp(n);
            disp(n~= floor(n));
                if isempty(equation) || ~isscalar(y0) || ~isfinite(y0) || ~isscalar(h) ||
~isfinite(h) || h<=0 || ~isscalar(n) || ~isfinite(n) || n<1 || n~=floor(n)
                    uialert(app.UIFigure,"Enter a valid equation, initial y, positive step size
and positive whole number of steps.","Input Error");
                    return;
                end
```

```
        f = str2func(['@(x,y)' equation]);
        xStart = 0;
        x = xStart;
        y = y0;
        data = zeros(n+1,4);
        data(1,:) = [0,x,y,NaN];
        for i = 1:n
          k1 = f(x,y);
          k2 = f(x+h/2,y+(h/2)*k1);
          k3 = f(x+h/2,y+(h/2)*k2);
          k4 = f(x+h/2,y+(h/2)*k3);
          yNew = y+(h/6)*(k1+2*k2+2*k3+k4);
          xNew = x+h;
          errorValue = abs(yNew -y);
          data(i+1,:) = [i,xNew,yNew,errorValue];
          x = xNew;
          y = yNew;
        end
        graphX = data(:,2);
        graphY = data(:,3);
        graphTitle = "Runge-Kutta Fourth Order Method";
        xLabelText = "x";
        yLabelText = "y";
        finalMessage = sprintf("RK4 calculation completed successfully.\nfinal
x = %.8f\nFinal y = %.8f\nSteps = %d",x,y,n);
    end
```

```
            app.UITable.Data = data;
            app.UITable.ColumnName = {"Iteration","x","Function Value/y","Error"};
            cla(app.UIAxes);
            plot(app.UIAxes,graphX,graphY,'-O',"LineWidth",1.5,"MarkerSize",5);
            xlabel(app.UIAxes,xLabelText);
            ylabel(app.UIAxes,yLabelText);
            title(app.UIAxes,graphTitle);
            grid(app.UIAxes,"on");
            uialert(app.UIFigure,finalMessage,"Calculation Results");
        catch ME
            uialert(app.UIFigure,ME.message,"Calculation Error");
    end
end

    % Button pushed function: ClearresultsButton
    function ClearresultsButtonPushed(app, event)
app.EnterfxorfxyEditField.Value = "";
        app.DerivativefxEditField_2.Value = "";
        app.FirstinitialguessEditField.Value = 0;
        app.SecondguessforSecantEditField.Value = 0;
        app.InitialyforRK4EditField.Value = 0;
        app.StepsizehEditField.Value = 0.1;
        app.NumberofStepsEditField.Value = 10;
        app.StoppingtoleranceEditField.Value = 0.000001;
        app.MaximumiterationEditField.Value = 100;
        app.UITable.Data = [];
```

```
        cla(app.UIAxes);
        title(app.UIAxes,"");
        xlabel(app.UIAxes,"");
        ylabel(app.UIAxes,"");
    end
end

% Component initialization
methods (Access = private)

    % Create UIFigure and components
    function createComponents(app)

        % Create UIFigure and hide until all components are created
        app.UIFigure = uifigure('Visible', 'off');
        app.UIFigure.Position = [100 100 900 650];
        app.UIFigure.Name = 'MATLAB App';

        % Create UIAxes
        app.UIAxes = uiaxes(app.UIFigure);
        title(app.UIAxes, 'ResultsAxes')
        xlabel(app.UIAxes, 'X')
        ylabel(app.UIAxes, 'Y')
        zlabel(app.UIAxes, 'Z')
        app.UIAxes.Position = [366 50 482 286];
```

```
        % Create NumericalMethodsSolverLabel
        app.NumericalMethodsSolverLabel = uilabel(app.UIFigure);
        app.NumericalMethodsSolverLabel.FontSize = 36;
        app.NumericalMethodsSolverLabel.FontWeight = 'bold';
        app.NumericalMethodsSolverLabel.Position = [104 594 466 47];
        app.NumericalMethodsSolverLabel.Text = 'Numerical Methods Solver';

        % Create SelectmethodDropDownLabel
        app.SelectmethodDropDownLabel = uilabel(app.UIFigure);
        app.SelectmethodDropDownLabel.HorizontalAlignment = 'right';
        app.SelectmethodDropDownLabel.Position = [123 528 82 22];
        app.SelectmethodDropDownLabel.Text = 'Select method';

        % Create SelectmethodDropDown
        app.SelectmethodDropDown = uidropdown(app.UIFigure);
        app.SelectmethodDropDown.Items = {'Newton-Raphson', 'Secant', 'Runge-Kutta',
'Option 4'};
        app.SelectmethodDropDown.Position = [220 528 100 22];
        app.SelectmethodDropDown.Value = 'Newton-Raphson';

        % Create FirstinitialguessEditFieldLabel
        app.FirstinitialguessEditFieldLabel = uilabel(app.UIFigure);
        app.FirstinitialguessEditFieldLabel.HorizontalAlignment = 'right';
        app.FirstinitialguessEditFieldLabel.Position = [113 416 94 22];
        app.FirstinitialguessEditFieldLabel.Text = 'First initial guess';
```

```
% Create FirstinitialguessEditField
app.FirstinitialguessEditField = uieditfield(app.UIFigure, 'numeric');
app.FirstinitialguessEditField.Position = [222 416 100 22];

% Create SecondguessforSecantEditFieldLabel
app.SecondguessforSecantEditFieldLabel = uilabel(app.UIFigure);
app.SecondguessforSecantEditFieldLabel.HorizontalAlignment = 'right';
app.SecondguessforSecantEditFieldLabel.Position = [68 379 139 22];
app.SecondguessforSecantEditFieldLabel.Text = 'Second guess for Secant';

% Create SecondguessforSecantEditField
app.SecondguessforSecantEditField = uieditfield(app.UIFigure, 'numeric');
app.SecondguessforSecantEditField.Position = [222 379 100 22];

% Create InitialyforRK4EditFieldLabel
app.InitialyforRK4EditFieldLabel = uilabel(app.UIFigure);
app.InitialyforRK4EditFieldLabel.HorizontalAlignment = 'right';
app.InitialyforRK4EditFieldLabel.Position = [121 344 86 22];
app.InitialyforRK4EditFieldLabel.Text = 'Initial y for RK4';

% Create InitialyforRK4EditField
app.InitialyforRK4EditField = uieditfield(app.UIFigure, 'numeric');
app.InitialyforRK4EditField.Position = [222 344 100 22];

% Create StepsizehEditFieldLabel
app.StepsizehEditFieldLabel = uilabel(app.UIFigure);
```

```
app.StepsizehEditFieldLabel.HorizontalAlignment = 'right';
app.StepsizehEditFieldLabel.Position = [143 303 64 22];
app.StepsizehEditFieldLabel.Text = 'Step size h';

% Create StepsizehEditField
app.StepsizehEditField = uieditfield(app.UIFigure, 'numeric');
app.StepsizehEditField.Position = [222 303 100 22];

% Create StoppingtoleranceEditFieldLabel
app.StoppingtoleranceEditFieldLabel = uilabel(app.UIFigure);
app.StoppingtoleranceEditFieldLabel.HorizontalAlignment = 'right';
app.StoppingtoleranceEditFieldLabel.Position = [103 211 105 22];
app.StoppingtoleranceEditFieldLabel.Text = 'Stopping tolerance';

% Create StoppingtoleranceEditField
app.StoppingtoleranceEditField = uieditfield(app.UIFigure, 'numeric');
app.StoppingtoleranceEditField.Position = [223 211 100 22];

% Create MaximumiterationEditFieldLabel
app.MaximumiterationEditFieldLabel = uilabel(app.UIFigure);
app.MaximumiterationEditFieldLabel.HorizontalAlignment = 'right';
app.MaximumiterationEditFieldLabel.Position = [105 172 103 22];
app.MaximumiterationEditFieldLabel.Text = 'Maximum iteration';

% Create MaximumiterationEditField
app.MaximumiterationEditField = uieditfield(app.UIFigure, 'numeric');
```

```
        app.MaximumiterationEditField.Position = [223 172 100 22];

        % Create CalculateButton
        app.CalculateButton = uibutton(app.UIFigure, 'push');
        app.CalculateButton.ButtonPushedFcn = createCallbackFcn(app,
@CalculateButtonPushed, true);
        app.CalculateButton.Position = [102 120 100 22];
        app.CalculateButton.Text = 'Calculate';

        % Create ClearresultsButton
        app.ClearresultsButton = uibutton(app.UIFigure, 'push');
        app.ClearresultsButton.ButtonPushedFcn = createCallbackFcn(app,
@ClearresultsButtonPushed, true);
        app.ClearresultsButton.Position = [222 120 100 22];
        app.ClearresultsButton.Text = 'Clear results';

        % Create UITable
        app.UITable = uitable(app.UIFigure);
        app.UITable.ColumnName = {'''Iteration'''; '''x or t'''; '''Approximation y'''; '''Error'''};
        app.UITable.RowName = {};
        app.UITable.Position = [410 344 425 185];

        % Create EnterfxorfxyEditFieldLabel
        app.EnterfxorfxyEditFieldLabel = uilabel(app.UIFigure);
        app.EnterfxorfxyEditFieldLabel.HorizontalAlignment = 'right';
        app.EnterfxorfxyEditFieldLabel.Position = [109 493 98 22];
        app.EnterfxorfxyEditFieldLabel.Text = 'Enter f(x) or f(x,y)';
```

```
% Create EnterfxorfxyEditField
app.EnterfxorfxyEditField = uieditfield(app.UIFigure, 'text');
app.EnterfxorfxyEditField.Position = [222 493 100 22];

% Create DerivativefxEditFieldLabel
app.DerivativefxEditFieldLabel = uilabel(app.UIFigure);
app.DerivativefxEditFieldLabel.HorizontalAlignment = 'right';
app.DerivativefxEditFieldLabel.Position = [126 -48 81 22];
app.DerivativefxEditFieldLabel.Text = 'Derivative f''(x)';

% Create DerivativefxEditField
app.DerivativefxEditField = uieditfield(app.UIFigure, 'text');
app.DerivativefxEditField.Position = [222 -48 100 22];

% Create DerivativefxEditField_2Label
app.DerivativefxEditField_2Label = uilabel(app.UIFigure);
app.DerivativefxEditField_2Label.HorizontalAlignment = 'right';
app.DerivativefxEditField_2Label.Position = [126 459 81 22];
app.DerivativefxEditField_2Label.Text = 'Derivative f''(x)';

% Create DerivativefxEditField_2
app.DerivativefxEditField_2 = uieditfield(app.UIFigure, 'text');
app.DerivativefxEditField_2.Position = [222 459 100 22];

% Create ResultsTableLabel
```

```
        app.ResultsTableLabel = uilabel(app.UIFigure);
        app.ResultsTableLabel.FontSize = 14;
        app.ResultsTableLabel.FontWeight = 'bold';
        app.ResultsTableLabel.Position = [544 540 91 22];
        app.ResultsTableLabel.Text = 'ResultsTable';

        % Create NumberofStepsEditFieldLabel
        app.NumberofStepsEditFieldLabel = uilabel(app.UIFigure);
        app.NumberofStepsEditFieldLabel.HorizontalAlignment = 'right';
        app.NumberofStepsEditFieldLabel.Position = [112 260 95 22];
        app.NumberofStepsEditFieldLabel.Text = 'Number of Steps';

        % Create NumberofStepsEditField
        app.NumberofStepsEditField = uieditfield(app.UIFigure, 'numeric');
        app.NumberofStepsEditField.Position = [222 260 100 22];

        % Show the figure after all components are created
        app.UIFigure.Visible = 'on';
    end
end

% App creation and deletion
methods (Access = public)

    % Construct app
    function app = NumericalMethodsSolver_Backup
```

```
        % Create UIFigure and components
        createComponents(app)

        % Register the app with App Designer
        registerApp(app, app.UIFigure)

        if nargout == 0
            clear app
        end
    end

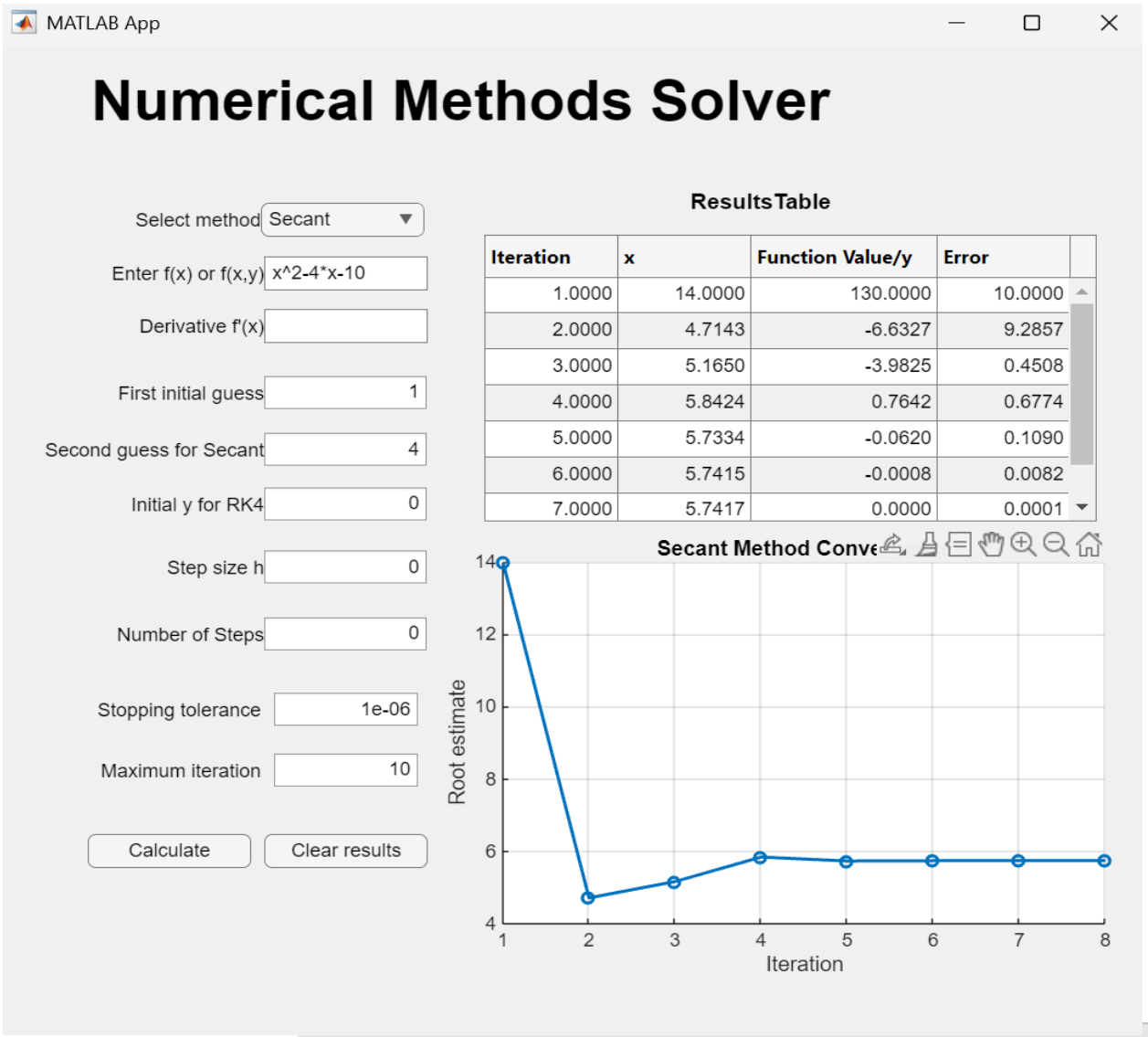
    % Code that executes before app deletion
    function delete(app)

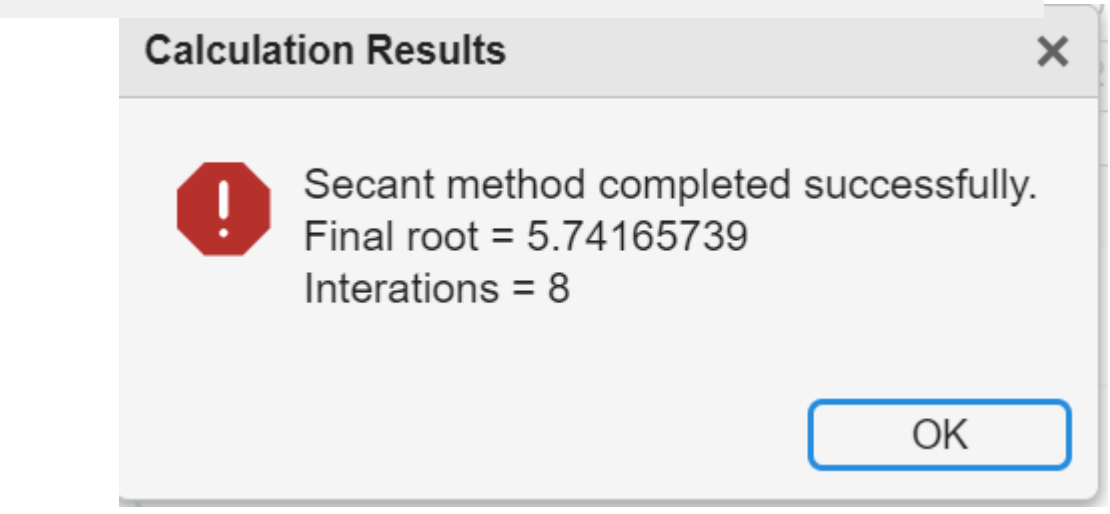
        % Delete UIFigure when app is deleted
        delete(app.UIFigure)
    end
  end
end
```

# SECANT METHOD

Find the root of equation $f(x) = x^2 - 4x - 10$ that lies between 1 and 4?

**OUTPUT:**

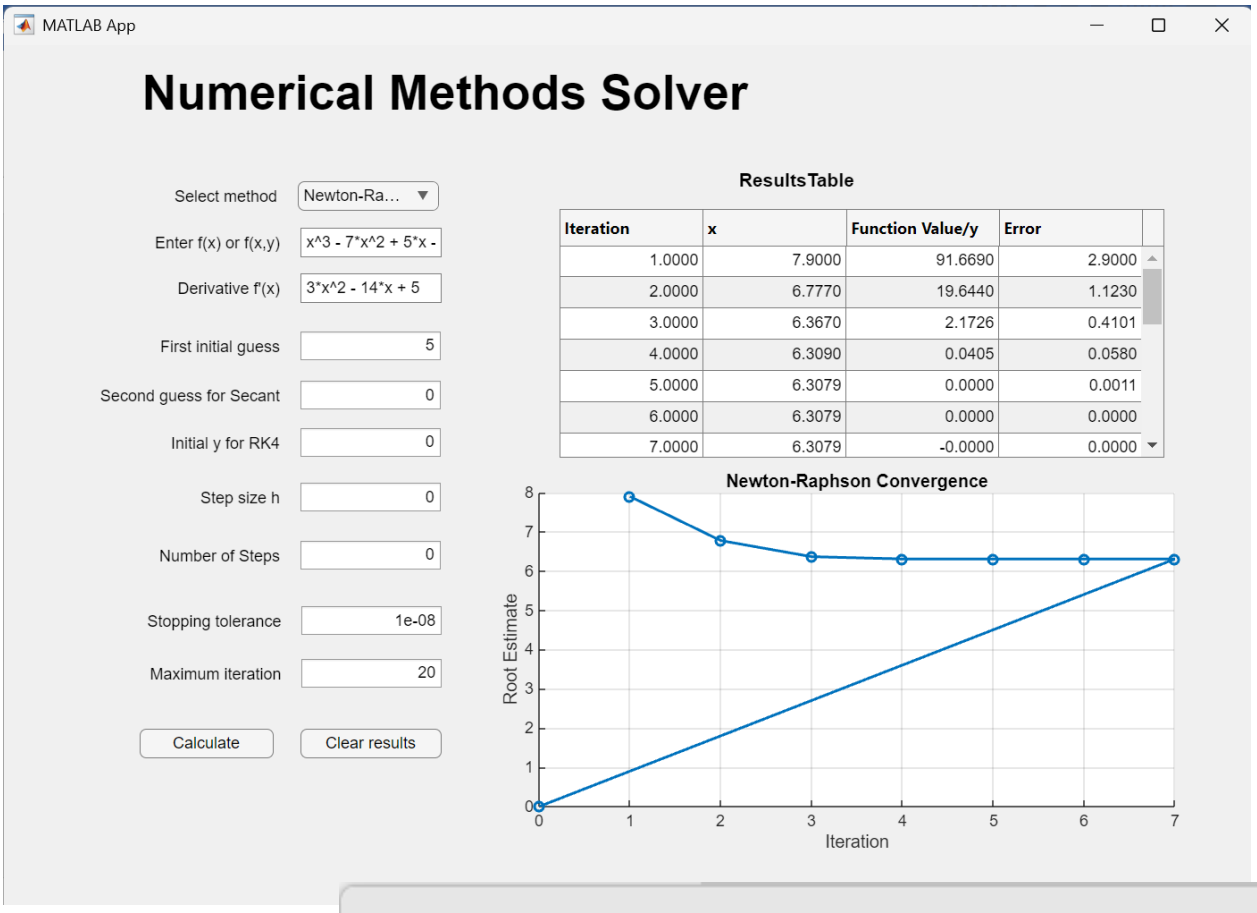


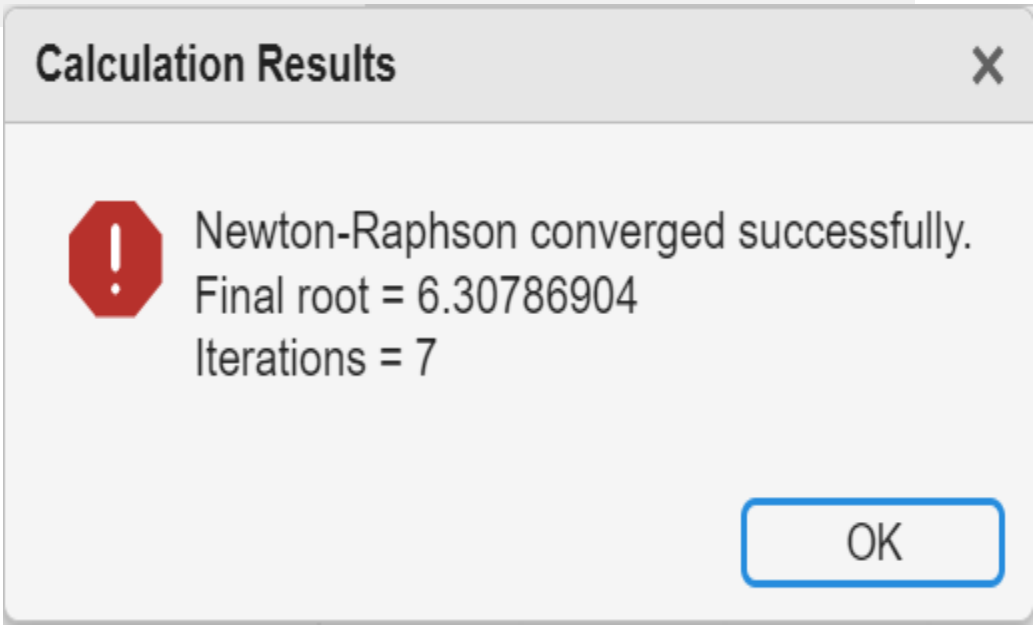



# NEWTON RAHPSON METHOD

Use Newton-Raphson method to solve the equation $x^3 - 7x^2 + 5x - 4 = 0$ near $x_0 = 5$ correct to 8 decimal places.

**OUTPUT:**



## RUNGE KUTTA METHOD

### Given the Initial Value Problem (IVP):

$$\frac{d_y}{d_x} = x + y, y(0) = 1$$

Using the **4**th Order Runge-Kutta Method, approximate the value of $y\ at\ x\ =\ 0.2$ taking a step size $of\ h\ =\ 0.1$

## OUTPUT:

